ASSESSMENT TOOL II- ASSIGNMENT

COURSE CODE: ITA1410 **COURSE NAME: Ethical Hacking**

COURSE OUTCOME COVERED

CO1: Apply the principle of the foundational concepts of ethical hacking, including types of hackers, ethical principles, legal frameworks, and the role of responsible disclosure in cybersecurity.

ASSIGNMENTS QUESTIONS

1. An AI-powered bot automatically probes IoT devices for default passwords and reports vulnerable devices to a public database without owner consent. Analyze whether the creator of this bot fits any traditional hacker classification, and what new categories might be needed.

Code of Conduct:

I PRAHARSHINI(192424085) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Rubrics

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all data-handling requirements and operations	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate data structures and algorithms to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct code, output and complexity analysis with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

1. Introduction

Ethical hacking plays an important role in protecting computer systems, networks, and digital devices from cyberattacks. It involves identifying security weaknesses and helping organizations improve their security. However, hacking activities must be performed within proper legal and ethical boundaries.

The Internet of Things (IoT) refers to physical devices connected to the internet, such as smart cameras, home appliances, smart TVs, routers, and industrial sensors. Many IoT devices are vulnerable to cyberattacks when users continue to use default usernames and passwords instead of changing them.

In the given scenario, an AI-powered bot automatically checks IoT devices for default passwords and publishes the details of vulnerable devices in a public database without obtaining permission from their owners. Although the creator may claim that the purpose is to improve security, the method raises important questions about authorization, privacy, responsibility, and the possible misuse of the information.

This situation shows that the traditional classification of hackers may not always be sufficient to explain modern cybersecurity activities involving artificial intelligence and automation.

2. Understanding the Given Scenario

The scenario contains four important elements.

1. Use of artificial intelligence: The bot automatically identifies and checks devices, reducing the need for manual testing.

2. Detection of default passwords: It searches for devices that may still use factory-set credentials, which can make them easier for attackers to access.

3. Lack of owner consent: The device owners have not authorized the scanning activity. This is a major ethical concern because security testing should be conducted within an approved scope.

4. Public disclosure of vulnerabilities: The bot publishes information about vulnerable devices in a public database. This may help security researchers, but it can also help malicious individuals locate and target those devices.

Therefore, the main issue is not simply whether the bot finds real vulnerabilities. The important question is whether the creator has the authority to perform the testing and whether the information is handled responsibly.

3. Traditional Types of Hackers

Hackers are commonly classified into three main categories: white-hat hackers, black-hat hackers, and grey-hat hackers. Other categories, such as script kiddies and hacktivists, may also be relevant depending on the circumstances.

3.1 White-Hat Hacker

A white-hat hacker is a security professional who identifies vulnerabilities with proper authorization. These hackers usually work with organizations to find weaknesses before attackers can exploit them.

Characteristics:

- Obtains permission before conducting security tests.
- Works within the agreed testing scope.
- Protects confidential information.
- Reports vulnerabilities to the responsible organization.
- Follows legal and professional standards.

Application to the scenario:

The creator's bot may have a security-improvement objective, but the absence of owner consent means that the activity does not meet the normal requirements of authorized white-hat testing. Good intentions alone are not enough to classify someone as an ethical hacker.

3.2 Black-Hat Hacker

A black-hat hacker performs unauthorized activities for malicious purposes, such as stealing information, compromising devices, causing damage, or gaining an unlawful advantage.

Characteristics:

- Accesses systems without authorization.
- May exploit security weaknesses for personal benefit.
- Can expose private information.
- May cause financial, operational, or reputational damage.
- Often ignores ethical and legal boundaries.

Application to the scenario:

If the creator uses the bot to identify targets for attacks, steal information, or intentionally expose device owners to harm, the creator may fit the black-hat classification. However, the lack of consent by itself does not prove malicious intent. The creator's actions and their consequences must also be examined.

3.3 Grey-Hat Hacker

A grey-hat hacker operates between the traditional white-hat and black-hat categories. Such a person may discover vulnerabilities without permission but may not have the intention to cause harm.

Characteristics:

- May conduct security testing without prior authorization.
- May identify genuine security weaknesses.
- May notify owners after discovering vulnerabilities.
- Can create risks even when the intention is to improve security.
- May cross legal or ethical boundaries.

Application to the scenario:

If the creator develops the bot to improve IoT security and does not intend to harm anyone, the activity may resemble grey-hat hacking. However, publishing vulnerable devices publicly without first notifying their owners may increase the risk of exploitation. Therefore, the creator cannot be considered fully ethical merely because the stated intention is positive.

3.4 Script Kiddie

A script kiddie is generally a person who uses existing hacking tools or scripts without fully understanding their technical operation.

Application to the scenario:

If the creator simply uses an existing AI scanning tool without understanding how it works, its limitations, or the risks it creates, some aspects of the activity may resemble script-kiddie behaviour. However, developing a sophisticated bot does not automatically make someone a script kiddie. The classification depends on the person's actual technical understanding and actions.

3.5 Hacktivist

A hacktivist uses hacking or digital activities to promote a political, social, or ideological cause.

Application to the scenario:

If the creator publishes vulnerable IoT devices to pressure manufacturers into improving security or to raise public awareness, the activity might have a hacktivist motivation. Nevertheless, a social or political objective does not remove the requirement to respect authorization, privacy, and applicable laws.

4. Comparison of Traditional Hacker Classifications

Hacker type	Main intention	Authorization	Relevance to the scenario

White-hat	Improve security	Normally required	Does not fit the normal authorized-testing model
Black-hat	Cause harm or gain an unlawful advantage	Absent	May fit if the creator acts maliciously
Grey-hat	Discover weaknesses, potentially without harmful intent	May be absent	May be the closest fit if the intention is security improvement
Script kiddie	Use tools without sufficient technical understanding	Depends on the activity	Relevant only if the creator lacks sufficient understanding
Hacktivist	Promote a cause or ideology	May be absent	Relevant if public disclosure is intended as activism

The comparison demonstrates that the creator cannot be classified solely by the technology used. Intention, authorization, technical understanding, and actual impact must all be considered.

5. Why Traditional Classifications May Not Be Sufficient

Traditional hacker classifications were largely developed around human decisions and actions. AI-powered systems introduce additional challenges because they can operate automatically, at high speed, and across many devices.

5.1 High-speed automated scanning

A human researcher may check a limited number of devices, whereas an automated bot can potentially scan a large number of devices in a short period. This increases the possible scale of both security benefits and harm.

5.2 Limited human supervision

An AI-powered bot may continue scanning and publishing information without a person reviewing each result. If the system is poorly designed, it may disclose sensitive details or mistakenly identify a device as vulnerable.

5.3 Difficulty in assigning responsibility

When a bot causes harm, responsibility may involve the developer, operator, organization deploying it, or other parties. The use of AI does not remove human accountability, but it can make responsibility more complicated to investigate.

5.4 Dual-use technology

A tool designed to protect IoT devices may also help attackers identify vulnerable targets. The same technical capability can be used for defensive or harmful purposes depending on its design, permissions, and operation.

5.5 Public disclosure risks

Publishing vulnerabilities may increase awareness and encourage manufacturers to fix security problems. However, it can also give attackers a list of vulnerable devices before the owners have secured them.

These challenges suggest that modern hacker classification should consider not only a person's intention but also the degree of automation, the system's level of autonomy, the authorization model, and the way discovered information is handled.

6. Proposed New Categories for AI-Based Hacking Activities

The following categories are proposed ways of describing AI-enabled cybersecurity activities. They are not universally recognized official hacker classifications.

6.1 AI-Assisted Ethical Hacker

An AI-assisted ethical hacker uses artificial intelligence to identify and analyze security weaknesses within an authorized environment.

Example: A cybersecurity professional uses an AI tool to scan a company's IoT devices after receiving written permission and submits a private report to the company.

This category highlights the use of AI while preserving the normal requirements of ethical hacking.

6.2 Autonomous Vulnerability Scanner Operator

This category describes a person who develops or operates a system that automatically searches for security weaknesses across many devices.

Example: A developer creates a bot that scans registered devices belonging to an organization and automatically generates security reports.

The category focuses on automation and the operator's responsibility for setting limits, monitoring results, and controlling data disclosure. It does not automatically indicate whether the operator is ethical.

6.3 AI-Enabled Grey-Hat Hacker

An AI-enabled grey-hat hacker uses AI tools to discover vulnerabilities without prior permission, potentially with the intention of improving security.

Example: A person creates a bot that discovers default passwords on internet-connected cameras and publishes the results to encourage users to change them.

This closely resembles the given scenario if the creator intends to improve security but has not obtained authorization. The activity may still be unethical or unlawful, depending on the actions and applicable law.

6.4 Malicious AI Operator

A malicious AI operator uses autonomous tools to identify or exploit vulnerabilities for harmful purposes.

Example: A person uses an automated system to locate insecure IoT devices so that they can be compromised or used in further attacks.

This category emphasizes the harmful purpose of using AI rather than simply the presence of automation.

6.5 Responsible Disclosure Researcher

A responsible disclosure researcher discovers vulnerabilities and reports them through appropriate private channels so that affected parties can investigate and fix the problem.

Example: A researcher discovers that a group of IoT cameras uses weak default credentials, privately informs the manufacturer, and allows a reasonable period for remediation before considering public disclosure.

This category emphasizes how information is handled after discovery. However, responsible disclosure does not automatically make an unauthorized method of discovery lawful.

6.6 AI Security Auditor

An AI security auditor uses automated systems to assess whether devices, applications, or networks follow established security requirements.

Example: An organization authorizes an auditor to evaluate its smart sensors and checks whether default passwords have been changed.

The auditor's work is based on a defined scope, documented procedures, evidence collection, and recommendations for improvement**.**

7. Comparison of Proposed AI-Related Categories

Proposed category	Main purpose	Permission status	Main responsibility
AI-assisted ethical hacker	Improve security	Authorized	Conduct safe testing
Autonomous scanner operator	Automate vulnerability discovery	Depends on deployment	Control scanning and disclosure
AI-enabled grey-hat hacker	Discover weaknesses without prior permission	Often unauthorized	Avoid harm and respect legal boundaries
Malicious AI operator	Support harmful activities	Unauthorized or abusive	Account for harmful actions

Responsible disclosure researcher	Help affected parties fix vulnerabilities	Depends on discovery method	Protect findings and notify the appropriate parties
AI security auditor	Evaluate security compliance	Authorized	Produce accurate and confidential reports

These categories can overlap. For example, an AI-assisted ethical hacker may also be a security auditor. The categories should therefore be treated as descriptive labels rather than rigid legal classifications.

8. Ethical Principles Involved

The scenario involves several important principles of ethical hacking.

8.1 Authorization

Permission should be obtained before testing devices owned by other people or organizations. The testing scope should clearly identify which systems can be assessed and which actions are allowed.

8.2 Non-maleficence

Non-maleficence means avoiding unnecessary harm. The creator should ensure that the bot does not disrupt devices, expose personal information, or create additional risks for device owners.

8.3 Confidentiality

Security findings may contain sensitive information. Publishing device addresses, credentials, or detailed vulnerability information can expose owners to further attacks. Such information should be protected and shared only with appropriate parties.

8.4 Accountability

The developer or operator must take responsibility for the bot's behaviour, including its scanning limits, data storage, disclosure decisions, and possible harmful outcomes.

8.5 Transparency

The purpose of the scanning activity, the type of information collected, and the process for reporting vulnerabilities should be clearly documented. Transparency helps affected parties understand and respond to security risks.

8.6 Responsible disclosure

Responsible disclosure generally involves notifying the affected owner, manufacturer, or security team privately and coordinating a reasonable opportunity to fix the vulnerability before public disclosure.

8.7 Proportionality

The method used to discover a vulnerability should be proportionate to the security objective. Testing should avoid unnecessary access, collection of personal data, or disruption of services.

9. Legal Framework and Responsibility

Cybersecurity activities must follow the laws applicable to the systems being tested and the locations involved. A claim that the bot is being used for public benefit does not automatically provide legal permission to scan devices.

In India, the Information Technology Act, 2000, including Section 43 and, where its requirements are met, Section 66, may be relevant to unauthorized access or related computer activities. The exact legal consequences depend on the facts and the applicable provisions. Other laws may also be relevant if personal data or privacy is involved.

The creator should therefore consider the following precautions:

1. Obtain written authorization from device owners before scanning.
2. Define the permitted devices, testing methods, and duration.
3. Avoid accessing or collecting unnecessary personal information.
4. Keep vulnerability reports private until an appropriate disclosure process is followed.
5. Maintain records of authorization, scanning activity, and disclosure decisions.
6. Stop scanning when permission is withdrawn or the system operates outside its approved scope.

The legal classification should not be decided from the scenario alone. The exact conduct, available evidence, intent, and applicable law must be considered.

10. Role of Responsible Disclosure in the Scenario

Responsible disclosure is particularly important because the bot publishes information in a public database before the device owners have given consent or had an opportunity to fix the issue.

A safer process would be:

Step 1: Authorized testing

The bot should scan only devices that are explicitly included in an approved testing scope.

Step 2: Verification

The operator should check that a finding is accurate without performing unnecessary access or collecting sensitive information.

Step 3: Private notification

The operator should contact the device owner, manufacturer, or appropriate security response team through an established reporting channel.

Step 4: Remediation opportunity

The affected party should receive a reasonable opportunity to investigate and fix the weakness.

Step 5: Coordinated public disclosure

If public disclosure is appropriate, the information should be released in a way that minimizes the risk of further attacks and follows applicable disclosure policies.

Step 6: Follow-up

The operator should verify the resolution only when authorized and update the report accordingly.

This process supports security improvement while reducing the likelihood that the disclosure itself will harm the people it is intended to protect.

11. Advantages and Risks of the AI-Powered Bot

Advantages

- Can identify common security weaknesses more quickly than manual inspection.
- Can help organizations discover widespread use of default credentials.
- Can support the development of better IoT security practices.
- Can generate reports that help prioritize security improvements.
- Can increase awareness of the importance of changing default passwords.

Risks

- May scan devices without permission.
- May expose vulnerable systems to attackers.
- May publish inaccurate or outdated findings.
- May collect more information than is necessary.
- May operate at a scale that increases the impact of errors.
- May create uncertainty about responsibility when automated decisions cause harm.

The benefits can be achieved more safely by using authorized scanning, privacy-preserving reporting, human oversight, and a clearly defined disclosure process.

12. Recommendations for Improving the Bot

The creator should redesign the bot around the following safeguards.

1. Permission-based scanning: Allow scanning only of registered or explicitly authorized devices.

2. Scope restrictions: Limit the bot to approved IP ranges, devices, and testing methods.

3. Human supervision: Require a responsible person to review significant findings before disclosure.

4. Privacy protection: Avoid storing passwords, personal information, or unnecessary device details.

5. Private reporting: Notify affected owners or security teams rather than immediately publishing sensitive findings.

6. Accurate verification: Reduce false positives and ensure reports contain sufficient evidence without exposing confidential information.

7. Audit records: Maintain records of permissions, scans, findings, and disclosure actions.

8. Emergency controls: Provide a way to stop the bot if it behaves unexpectedly or causes disruption.

9. Clear accountability: Identify who is responsible for operating the bot and handling reports.

10. Regular review: Evaluate the bot's security, performance, and compliance with applicable policies.

These recommendations would make the system more useful for legitimate security research while reducing its potential for misuse.

13. Critical Analysis

The most important question in this case is whether a good intention can justify unauthorized security testing. The answer is no. Finding a real vulnerability does not automatically make the method ethical.

The creator may believe that publishing vulnerable devices will encourage manufacturers and users to improve security. However, the public database may also provide attackers with information about systems that have not yet been secured. The activity could therefore create a greater risk than the creator intended.

At the same time, the absence of permission does not by itself prove that the creator is a black-hat hacker. The creator may have no intention of causing harm. This is why the grey-hat classification may be relevant, although it does not remove the ethical and legal concerns.

AI makes the situation more complex because a bot can repeat the same activity across a large number of devices. A single design error may lead to widespread exposure. For this reason, developers must consider both their own decisions and the foreseeable behaviour of the systems they create.

A better classification system would consider four dimensions: intention, authorization, level of automation, and impact. This approach provides a more complete understanding than assigning a label based only on whether the person claims to be helping others.

14. Final Conclusion

The creator of the AI-powered IoT scanning bot does not fit neatly into one traditional hacker classification. If the creator intends to improve security but performs testing without permission, the activity may resemble grey-hat hacking. If the creator deliberately uses the information to cause harm or facilitate attacks, the creator may fit the black-hat category. The creator does not meet the normal requirements of authorized white-hat testing merely by claiming a positive purpose.

The scenario also shows that AI changes the scale, speed, and complexity of cybersecurity activities. Proposed categories such as AI-assisted ethical hacker, autonomous vulnerability scanner operator, AI-enabled grey-hat hacker, and malicious AI operator can help describe these differences. However, these are suggested descriptive categories rather than universally accepted official classifications.

Ultimately, ethical hacking requires more than technical knowledge. It requires authorization, confidentiality, accountability, proportionality, and responsible disclosure. The creator should redesign the bot to scan authorized devices, protect sensitive information, notify affected parties privately, and provide a reasonable opportunity for remediation. In this way, AI can contribute to cybersecurity without unnecessarily exposing device owners to additional risks.